

B.5 Current and circuits

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **28 marks**

Show your working.

Question 1 — Fairy-light forensics

(6 marks)

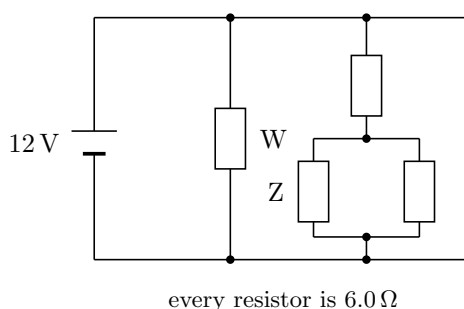
Two strings of fairy lights decorate a café: string S has its bulbs in series, string P in parallel, both run from identical supplies.

- (a) Define electric current, and state what an ammeter placed at different points around a single series loop would read. [3]
- (b) One bulb in each string blows (breaks its filament). Describe what happens to the rest of each string, with reasons. [2]
- (c) A customer suggests the last bulb in string S glows dimmest because “the current gets used up along the way”. State what is actually transferred, and what happens to the current. [1]

Question 2 — The current maze

(9 marks)

A 12 V cell of negligible internal resistance drives two parallel branches. Branch 1 is a single $6.0\ \Omega$ resistor, W. Branch 2 is a $6.0\ \Omega$ resistor in series with a parallel pair of $6.0\ \Omega$ resistors; Z is one of the pair.



- (a) Calculate the current in W. [2]
- (b) Calculate the current in Z. [3]
- (c) Hence write the ratio I_Z/I_W as a fraction. [1]
- (d) Calculate the total current from the cell and hence the resistance of the whole network. [3]

Question 3 — The stretched fence wire

(6 marks)

A farmer's fencing wire, resistance R , is accidentally stretched by a tractor to twice its original length. Its volume (and material) are unchanged.

- (a) State the equation for the resistance of a wire in terms of resistivity, length and cross-sectional area. [1]

- (b) Determine the new resistance in terms of R . [3]
- (c) The farmer reasons “twice the length, twice the resistance”. Identify the change the farmer has not accounted for. [2]

Question 4 — The tired battery (7 marks)

A worn 9.0 V (emf) battery has internal resistance $0.50\ \Omega$. It is connected to a $4.0\ \Omega$ speaker.

- (a) Define the emf of a cell. [2]
- (b) Calculate the current in the circuit. [2]
- (c) Calculate the terminal potential difference, and explain why it is less than the emf. [3]

Answers

Question 1 — Fairy-light forensics

- (a) Current is the rate of flow of charge, $I = \Delta q / \Delta t$. An ammeter reads the *same* value at every point of a single series loop, because charge is conserved and has nowhere else to go.
- (b) String S (series): the break opens the only path, so *all* its bulbs go out. String P (parallel): only the blown bulb goes out, since each remaining bulb keeps its own path and still has the full supply p.d. across it.
- (c) What is transferred is *energy*, per unit charge, which is the p.d. across each bulb. The current is the same everywhere in the series loop; it is not used up.

Question 2 — The current maze

- (a) W sits alone across the cell, so $I_W = 12/6.0 = 2.0$ A.
- (b) In branch 2 the parallel pair is $6.0 \parallel 6.0 = 3.0 \Omega$, in series with 6.0Ω , giving 9.0Ω . The branch current is $12/9.0 = 1.33$ A, and it splits equally between two identical resistors, so $I_Z = 0.67$ A.
- (c) $I_Z/I_W = \frac{2}{3}/2 = \frac{1}{3}$.
- (d) $I_{\text{total}} = 2.0 + 1.33 = 3.33$ A, so $R = 12/3.33 = 3.6 \Omega$. (Check: the two branches are 6.0Ω and 9.0Ω in parallel, and $\frac{6 \times 9}{6+9} = 3.6 \Omega$.)

Question 3 — The stretched fence wire

- (a) $R = \frac{\rho L}{A}$.
- (b) The length doubles, and since the volume LA is unchanged the cross-sectional area must halve. So $R' = \frac{\rho(2L)}{A/2} = 4\frac{\rho L}{A} = 4R$.
- (c) The cross-sectional area. Stretching necessarily thins the wire, and the two effects *multiply* rather than the length acting alone.

Question 4 — The tired battery

- (a) The work done per unit charge by the cell in driving charge around the complete circuit; equivalently, its terminal p.d. when no current flows.
- (b) $I = \frac{\varepsilon}{R+r} = \frac{9.0}{4.0+0.50} = \frac{9.0}{4.5} = 2.0$ A.
- (c) $V = \varepsilon - Ir = 9.0 - (2.0)(0.50) = 8.0$ V (equivalently $V = IR = 2.0 \times 4.0$). It falls short of the emf because one volt per coulomb is dissipated *inside* the battery, across its own internal resistance.